

PPA-6A — SCFSG CPU Architecture & Instruction Set

Title

Processor Architecture and Instruction Set Based on Symmetry-Constrained Finite-State Gates

Field of the Invention

The invention relates to processor architectures, instruction sets, and microarchitectures. More specifically, it relates to CPUs constructed from symmetry-constrained finite-state gates (SCFSGs) rather than Boolean combinational logic gates.

Background

Conventional CPUs are built from Boolean logic gates operating on independent bits, executing arithmetic and control instructions over linear registers and memory. Such architectures are poorly suited for computation in constrained state spaces where valid transitions are limited by symmetry, adjacency, or topology.

There is a need for a processor architecture whose primitive element is a finite-state gate with symmetry-constrained transitions, enabling computation as traversal of a constrained state graph rather than evaluation of arbitrary Boolean functions.

Summary

This invention discloses a **CPU architecture** composed of arrays of **Symmetry-Constrained Finite-State Gates (SCFSGs)**. Each SCFSG maintains an **8-bit state identifier** representing one of a finite set of valid states (e.g., 240 states). Instructions select one of a small, fixed set of **symmetry operators** (e.g., four operators encoded by two bits), which deterministically transition the state according to a predefined symmetry-constrained transition function.

The processor executes programs as **operator sequences**, optionally grouped into fixed-length **microinstruction blocks** (e.g., six-step blocks) to normalize parity, chirality, or other invariants. The architecture supports synchronous or asynchronous execution, pipelining, tiling, and clustering of SCFSG cores.

Detailed Description

1. SCFSG Core

Each SCFSG comprises:

- an **8-bit state register** (`state_id`),
- a **transition mechanism** (e.g., LUT/ROM) implementing $\text{state_id_next} = f(\text{state_id}, \text{op})$,
- an **operator input** (`op`), encoded as two bits selecting one of four symmetry operators.

The transition function is total but constrained: only symmetry-allowed transitions exist.

2. Instruction Set Architecture (ISA)

In a preferred embodiment:

- **Instructions** are operator selections (`op`) applied per cycle or per microstep.
- **Programs** are sequences of operator codes.
- **Execution** advances the SCFSG state deterministically.

Optionally:

- instructions are grouped into **fixed-length microblocks** (e.g., six operators per block),
 - block boundaries define synchronization points for higher-level coordination.
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3. Microarchitecture

The CPU may include:

- arrays of SCFSG cores (tiles),
- pipelines executing microsteps,
- block-level synchronization logic,
- state readout and write-back paths,
- debug and trace interfaces exposing state evolution.

No arithmetic logic unit (ALU) is required; computation is realized through state transitions.

4. Memory and Registers

Memory may store:

- operator sequences (programs),
- state snapshots,
- metadata derived from states.

Registers correspond to SCFSG state registers rather than numeric values.

5. Embodiments

The architecture may be implemented as:

- FPGA prototypes,
 - ASIC designs,
 - emulation layers on conventional CPUs,
 - hybrid systems combining SCFSG cores with traditional processors.
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Advantages

- Deterministic computation in constrained state spaces
 - Inherent exclusion of invalid states
 - Naturally parallel execution
 - Compatibility with existing silicon fabrication
 - Suitable for symmetry-native computation
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Provisional Statement

This application claims priority for processor architectures and instruction sets based on symmetry-constrained finite-state gates. It is independent of, but conceptually related to, earlier disclosures of SCFSG logic elements and spatial computing substrates.